

1. Task Description

The objective of this assignment was to train a reinforcement learning agent to solve the **MiniGrid-DoorKey-6x6-v0** environment using a **Deep Q-Network (DQN)**.

The task requires to:

1. Explore the environment
2. Find and pick up the key
3. Unlock the door
4. Navigate to the goal tile

2. Methodology

The solution was implemented using a value-based reinforcement learning approach with **Deep Q-Learning**.

2.1 Architecture

The agent used a Convolutional Neural Network (CNN) to process the 7×7 RGB observations.

Input Shape

(3, 7, 7)

Forward Pass

Observations were normalized inside the network:

$$x = x / 255.0$$

This improved numerical stability during training.

2.2 Core Steps

1. Observe current state
2. Select action using epsilon-greedy policy
3. Execute action in environment
4. Store transition in replay memory
5. Sample mini-batch from memory
6. Compute target Q-values
7. Minimize temporal difference loss
8. Update target network periodically

Loss Function

Mean Squared Error (MSE) between predicted Q-values and target Q-values.

Optimizer

Adam optimizer was used.

2.3 Parameters Used

Parameter	Value
Initial Epsilon	1.0
Minimum Epsilon	0.1
Decay Schedule	Linear
Decay Steps	50000

Formula

$$\text{epsilon} = \max(\text{epsilon_lb}, \text{epsilon_ub} - \text{timesteps} / \text{epsilon_decay})$$

1. Correct Input Tensor Shape

The model required batched inputs during inference and ONNX export.

Correct shape:(1, 3, 7, 7)

2. Observation Normalization

Pixel inputs were divided by 255 inside the model.

3. Experience Replay

Replay memory stored past transitions and randomly sampled mini batches.

4. Target Network Updates

A separate target network was maintained and updated periodically.

Parameter used:

| Target Update Frequency | 500 steps |

3. Training Details

The following parameters were used in the notebook.

Parameter	Value
Number of Episodes	5000
Replay Buffer Size	50000
Mini-batch Size	64
Learning Rate	0.0005
Discount Factor (Gamma)	0.99
Initial Timesteps	0
Update Start After	500
Train Every	4 steps
Target Update Frequency	500
Tau	1e-3
Evaluation Interval	100 episodes
Number of Evaluation Episodes	10

Training Process

For each episode:

1. Reset environment
2. Run until termination
3. Store transitions
4. Update network every 4 steps after replay buffer warm-up
5. Evaluate periodically
6. Save checkpoints

4. Results

Observed Behavior

During early episodes, the reward was often zero because the environment requires multiple sequential tasks:

- locate key
- pick key
- open door
- reach goal

This makes random success unlikely.

After continued training, the agent began receiving non-zero rewards, indicating successful completion of the task in some episodes.

Performance Indicators

- Initial stage: mostly zero rewards
- Intermediate stage: occasional successful episodes
- Later stage: increasing consistency depending on training duration

Final Outcome

The model was successfully:

1. Trained using DQN
2. Exported to ONNX